THE MINISTRY OF SCIENCE AND HIGHER EDUCATION
OF THE RUSSIAN FEDERATION

ITMO UNIVERSITY

International Research and Educational Center
for Physics of Nanostructures

LAB 4
for the Course
"Discrete Control Systems"

on the topic:
Synthesis of the discrete stabilizing control algorithms

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HDU-ITMO Joint Institute 2025

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INTRODUCTION

0.1 Introduction

The purpose of this work is to synthesize discrete stabilizing control algorithms using state feedback methods. The study focuses on the general principles of designing discrete controllers that provide specified quality indicators in closed-loop systems.

The main objectives of this laboratory work are:

– To develop mathematical models of the control object in both continuous and discrete time domains
– To verify controllability, observability and stability properties of the discrete system
– To design time-optimal discrete reference models
– To synthesize linear state feedback controllers using canonical transformation methods
– To verify controller performance through simulation and analysis

The work follows a systematic approach including derivation of state-space models, discretization using matrix exponential methods, analysis of system properties, reference model design, and controller synthesis. The MATLAB/Simulink environment is utilized for numerical computations and system simulations.

0.2 Notations

Throughout this work, the following notations are used:

Table 1 — Table 1: Notations of symbols

Symbol	Explanation
x	State vector
y	Output variable
u	Control input
A, B, C	State-space matrices (continuous time)
A_d, B_d	State-space matrices (discrete time)
T	Sampling time
K_d	Linear state feedback matrix
F_d	Closed-loop system matrix
U_d	Controllability matrix
U_n	Observability matrix
Γ_d	Reference model dynamics matrix
H_d	Reference model output matrix
M	Transformation matrix

0.3 Variant

Table 2 — Table 2: Parameters for Variant 22

Type	k_1	a_0^1	T_1	ξ	k_2	a_0^2	T_2	T
4	5.75	0	5	0	1	0	1	1

1 SOLUTIONS

The purpose of the work: To synthesize discrete stabilizing control algorithms using state feedback methods and analyze system properties including controllability, observability, and stability.

1.1 Methodology

The synthesis methodology includes:

- Development of continuous state-space model for the given control object
- Discretization using matrix exponential methods with sampling time $T = 1$ s
- Analysis of controllability, observability and stability properties
- Design of time-optimal discrete reference models
- Synthesis of linear state feedback controllers using canonical transformation
- Verification through simulation of closed-loop system performance

1.2 Problem Statement

Consider the continuous control object with the following parameters:

- Sampling time $T = 1$ s

The objective is to:

- Design the continuous ”input-state-output” model
- Perform discretization and verify correctness
- Analyze system properties (controllability, observability, stability)
- Design time-optimal reference model
- Synthesize LSFB matrix using canonical transformation
- Verify controller performance through simulation

1.3 Continuous State-Space Model

The continuous state-space representation is given by:

$$\dot{x} = Ax + Bu \tag{1}$$

$$y = Cx \tag{2}$$

where:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1.1500 & 0 \end{bmatrix} \tag{3}$$

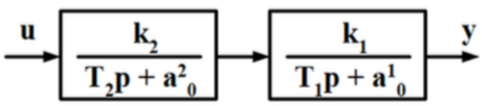

Figure 1 — Control object block diagram

1.4 Discrete State-Space Model

Using the sampling time $T = 1$ s, the discrete-time matrices are:

$$A_d = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B_d = \begin{bmatrix} 0.5000 \\ 1.0000 \end{bmatrix} \tag{4}$$

The discrete system is represented as:

$$x(k+1) = A_d x(k) + B_d u(k) \tag{5}$$

$$y(k) = Cx(k) \tag{6}$$

1.5 System Properties Analysis

1.5.1 Controllability Analysis

The controllability matrix is calculated as:

$$U_d = \begin{bmatrix} B_d & A_d B_d \end{bmatrix} = \begin{bmatrix} 0.5000 & 1.5000 \\ 1.0000 & 1.0000 \end{bmatrix} \tag{7}$$

The rank of U_d is 2, which equals the system order. Therefore, the system is **completely controllable**.

1.5.2 Observability Analysis

The observability matrix is calculated as:

$$U_n = \begin{bmatrix} C \\ CA_d \end{bmatrix} = \begin{bmatrix} 1.1500 & 0 \\ 1.1500 & 1.1500 \end{bmatrix} \tag{8}$$

The rank of U_n is 2, which equals the system order. Therefore, the system is **completely observable**.

1.5.3 Stability Analysis

The characteristic polynomial of the discrete system is:

$$D(z) = \det(zI - A_d) = z^2 - 2z + 1 \tag{9}$$

The eigenvalues are $z_1 = 1$ and $z_2 = 1$, both with magnitudes equal to 1. Therefore, the original system is **marginally stable**.

1.6 Time-Optimal Reference Model Design

For time-optimal control, all characteristic roots are placed at the origin:

$$z_1^* = z_2^* = 0 \tag{10}$$

The reference model is designed as:

$$x_r(k+1) = \Gamma_d x_r(k) \tag{11}$$

$$y_r(k) = H_d x_r(k) \tag{12}$$

where:

$$\Gamma_d = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad H_d = \begin{bmatrix} 1 & 0 \end{bmatrix} \tag{13}$$

The desired characteristic polynomial is:

$$D^*(z) = \det(zI - \Gamma_d) = z^2 \tag{14}$$

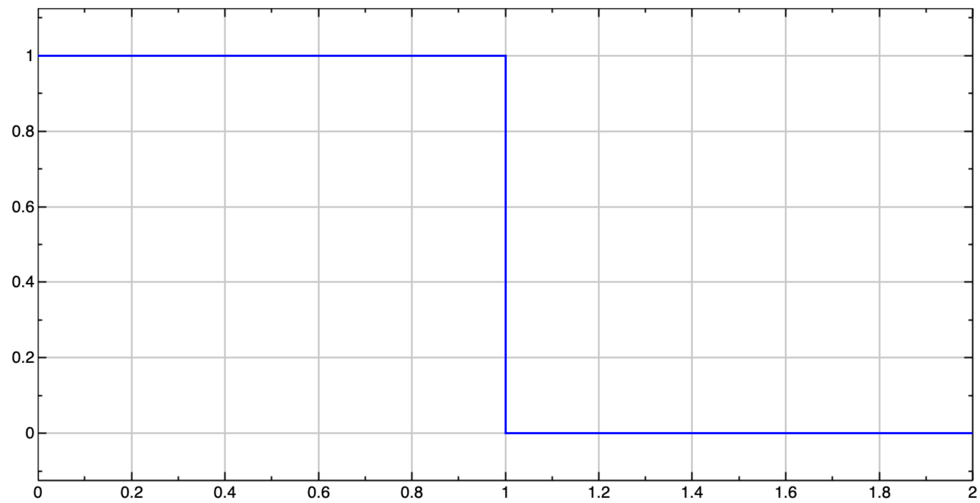

Figure 2 — Time-optimal reference model response

1.7 LSFB Matrix Synthesis

1.7.1 Canonical Transformation

The characteristic polynomial coefficients are:

$$a_i = \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} \tag{15}$$

The canonical system matrices are:

$$A_k = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}, \quad B_k = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{16}$$

The canonical controllability matrix is:

$$U_k = \begin{bmatrix} B_k & A_k B_k \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} \tag{17}$$

The transformation matrix is:

$$M = U_d U_k^{-1} = \begin{bmatrix} 0.5000 & 0.5000 \\ -1.0000 & 1.0000 \end{bmatrix} \tag{18}$$

1.7.2 Feedback Gains in Canonical Form

The desired characteristic polynomial coefficients are:

$$a_i^* = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \tag{19}$$

The feedback coefficients in canonical form are:

$$K^k = \begin{bmatrix} k_1^k & k_2^k \end{bmatrix} = \begin{bmatrix} -1 & 2 \end{bmatrix} \tag{20}$$

1.7.3 LSFB Matrix in Original Basis

The LSFB matrix in the original basis is:

$$K_d = K^k M^{-1} = \begin{bmatrix} 1.0000 & 1.5000 \end{bmatrix} \tag{21}$$

The closed-loop system matrix becomes:

$$F_d = A_d - B_d K_d = \begin{bmatrix} 0.5000 & 0.2500 \\ -1.0000 & -0.5000 \end{bmatrix} \tag{22}$$

1.7.4 Verification

The characteristic polynomial of the closed-loop system is:

$$\det(zI - F_d) = z^2 \tag{23}$$

The pole placement verification is successful - the closed-loop system has both eigenvalues at the origin, achieving the desired time-optimal behavior.

1.8 System Simulation and Analysis

1.8.1 Continuous vs Discrete Model Verification

The continuous and discrete models are simulated with step input signals to verify the correctness of discretization. The responses should match at sampling instants, confirming proper discretization.

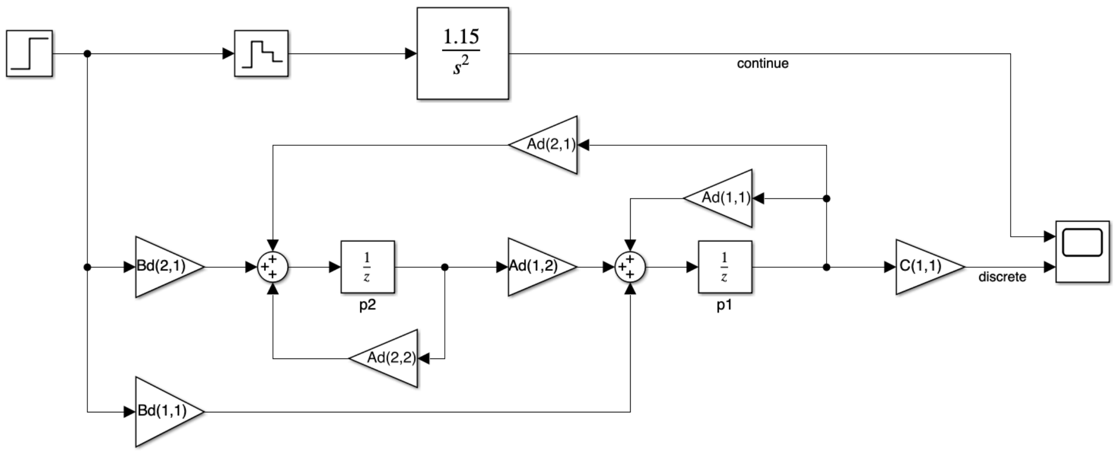

Figure 3 — Block diagram of the discrete control system

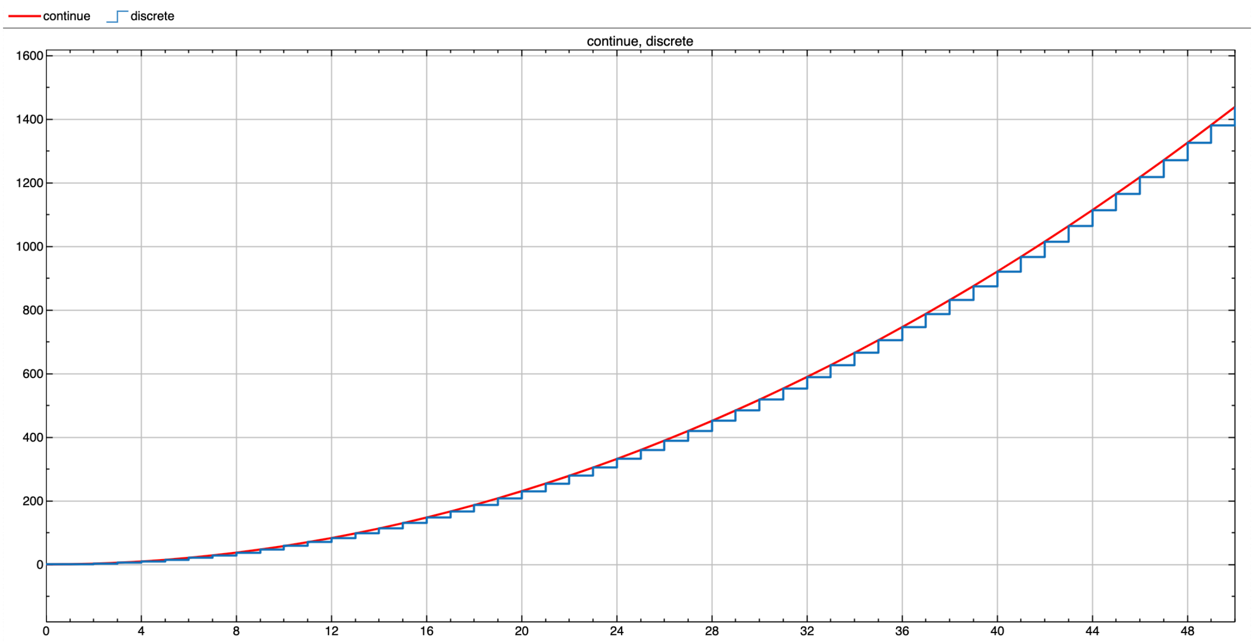

Figure 4 — Comparison of continuous and discrete system responses

1.8.2 Closed-Loop System Performance

The synthesized closed-loop system is simulated with initial conditions $y(0) = 1$, $\dot{y}(0) = 0$. Due to the successful pole placement, the system exhibits time-optimal behavior, reaching zero in minimum time.

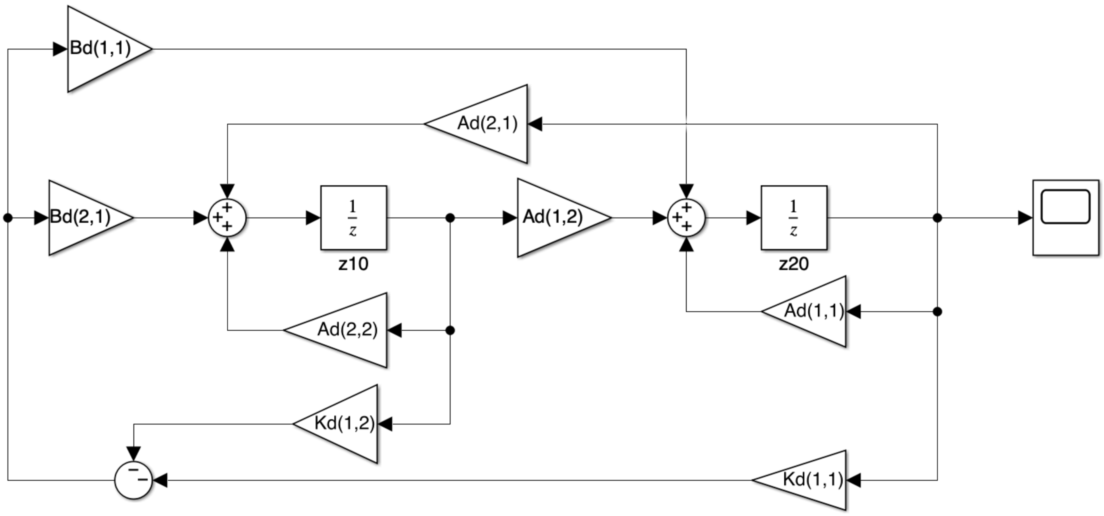

Figure 5 — Block diagram of the closed-loop control system

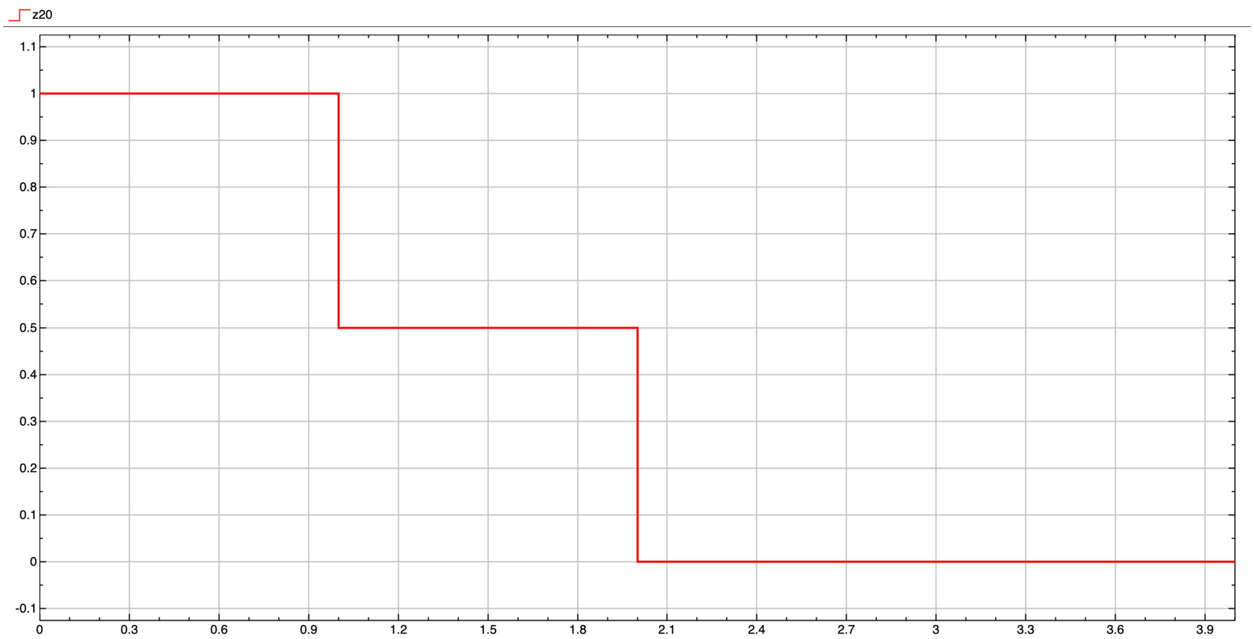

Figure 6 — Closed-loop system response with LSFB controller

1.9 Conclusion

The study investigated the synthesis of discrete stabilizing control algorithms using state feedback methods. Key findings include:

- The continuous control object was properly modeled with state-space matrices:
$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1.1500 & 0 \end{bmatrix}$$
- Discretization with sampling time $T = 1$ s resulted in:
$$A_d = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B_d = \begin{bmatrix} 0.5000 \\ 1.0000 \end{bmatrix}$$
- System analysis confirmed the system is **completely controllable** and **completely observable**, theoretically enabling effective state feedback control design
- Stability analysis showed the original system is **marginally stable** with eigenvalues $z_1 = 1$ and $z_2 = 1$
- Time-optimal reference model was designed with all characteristic roots at the origin, providing the theoretical fastest possible response
- LSFB matrix was successfully synthesized using canonical transformation methods, resulting in $K_d = \begin{bmatrix} 1.0000 & 1.5000 \end{bmatrix}$
- **Verification confirmed successful pole placement** - the closed-loop system characteristic polynomial is z^2 with both eigenvalues at the origin, achieving time-optimal control
- The closed-loop system exhibits the desired time-optimal behavior, reaching the equilibrium point in minimum time

The control synthesis was successful, demonstrating the effectiveness of the canonical transformation approach for discrete state feedback controller design. The system achieves the desired time-optimal performance with all poles placed at the origin.